

2021 IEEE International Conference on Blockchain (Blockchain) **Blockchain 2021**

Table of Contents

Message from the Steering Committee	xvi
Message from the General Chairs	xvii
Message from the Program Chairs	xviii
Organizing Committee	xix
Message from the Workshop on Blockchain Security, Application, and Performance	xxii
Message from the Workshop on Advances in Artificial Intelligence for Blockchain	xxiii
Message from the Symposium on Fintech and Blockchain Systems Chairs	xxiv
Message from the Blockchain & the Circular Supply Chain Workshop Organizers	xxv

The 4th IEEE International Conference on Blockchain (Blockchain-2021)

Regular Papers

Security and Attacks on Blockchain

Scalable Blockchain Anomaly Detection With Sketches	1
<i>Tomer Voronov (Technion), Danny Raz (Technion), and Ori Rottenstreich (Technion)</i>	
EtherProv: Provenance-Aware Detection, Analysis, and Mitigation of Ethereum Smart Contract Security Issues	11
<i>Shlomi Linoy (University of New Brunswick, Canada), Suprio Ray (University of New Brunswick, Canada), and Natalia Stakhanova (University of Saskatchewan, Canada)</i>	
Region-Based Neighbor Selection in Blockchain Networks	21
<i>Hiroshi Matsuura (NTT Network Service System Laboratories), Yoshinori Goto (NTT Network Service System Laboratories), and Hidehiro Sao (NTT Network Service System Laboratories)</i>	
Secure Remote Maintenance via Workflow-Driven Security Framework	29
<i>Prabhakaran Kasinathan (Cybersecurity Technology, Siemens AG, Germany), Davide Martintoni (Applied Research and Technology, Collins Aerospace, Italy), Benedikt Hofmann (Cybersecurity Technology, Siemens AG, Germany), Valerio Senni (Applied Research and Technology, Collins Aerospace, Italy), and Martin Wimmer (Cybersecurity Technology, Siemens AG, Germany)</i>	

FairTrade: Efficient Atomic Exchange-Based Fair Exchange Protocol for Digital Data Trading .38.....
Changhao Chenli (University of Notre Dame, USA), Wenyi Tang (University of Notre Dame, USA), and Taeho Jung (University of Notre Dame, USA)

System Applications and Performance

EIoVChain: Towards Authentication and Secure Communication Based Blockchain for Internet of Vehicles (IoV) .47.....
Amritesh Kumar (Indian Institute of Technology (IIT), India) and Debasis Das (Indian Institute of Technology (IIT), India)

ClaimChain: Secure Blockchain Platform for Handling Insurance Claims Processing .55.....
Naga Ramya Bhamidipati (University of Missouri-Columbia, USA; Jawaharlal Nehru Technological University, India), Varsha Vakkavanthula (University of Missouri-Columbia, USA; Jawaharlal Nehru Technological University, India), George Stafford (University of Missouri-Columbia, USA; Jawaharlal Nehru Technological University, India), Masrik Dahir (University of Missouri-Columbia, USA; Jawaharlal Nehru Technological University, India), Roshan Neupane (University of Missouri-Columbia, USA; Jawaharlal Nehru Technological University, India), Ernest Bonnah (University of Missouri-Columbia, USA; Jawaharlal Nehru Technological University, India), Songjie Wang (University of Missouri-Columbia, USA; Jawaharlal Nehru Technological University, India), J. V. R. Murthy (JNTU-Kakinada, USA; Jawaharlal Nehru Technological University, India), Khaza Anuarul Hoque (University of Missouri-Columbia, USA; Jawaharlal Nehru Technological University, India), and Prasad Calyam (University of Missouri-Columbia, USA; Jawaharlal Nehru Technological University, India)

CySCPro - Cyber Supply Chain Provenance Framework for Risk Management of Energy Delivery Systems .65.....
Eranga Bandara (Old Dominion University, USA), Deepak Tosh (University of Texas, USA), Sachin Shetty (Old Dominion University, USA), and Bheshaj Krishnappa (Reliability First, USA)

Irrationality, Extortion, or Trusted Third-Parties: Why It is Impossible to Buy and Sell Physical Goods Securely on the Blockchain .73.....
Amir Kafshdar Goharshady (The Hong Kong University of Science and Technology, Hong Kong)

Blockchain-Based Supply Chain System for Traceability, Regulation and Anti-Counterfeiting .82.....
Wang Fat Lau (The Hong Kong Polytechnic University, Hong Kong, China), Dennis Y. W. Liu (The Hong Kong Polytechnic University, Hong Kong, China), and Man Ho Au (The University of Hong Kong, Hong Kong, China)

Smart Contracts and Privacy

SmartBuilder: A Block-Based Visual Programming Framework for Smart Contract Development .90
Mpyana Mwamba Merlec (Korea University, South Korea), Youn Kyu Lee (Hongik University, South Korea), and Hoh Peter In (Korea University, South Korea)

Full-Stack Hierarchical Fusion of Static Features for Smart Contracts Vulnerability Detection .95.....	
<i>Wanqing Jie (Guangzhou University, China), Arthur Sandor Voundi Koe (Guangzhou University, China), Pengfei Huang (Guangzhou University, China), and Shiwen Zhang (Hunan University of Science and Technology, China)</i>	
BlockTorrent: A Blockchain Enabled Privacy-Preserving Data Availability Protocol for Multi-Stakeholder Scenarios .103.....	
<i>Ambrose Hill (University of New South Wales, Australia), Shailesh Mishra (IIT Kharagpur, India), Atharv Singh Patlan (IIT Kharagpur, India), Ali Dorri (Queensland University of Technology, Australia), Volkan Dedeoglu (CSIRO, Australia), Raja Jurdak (Queensland University of Technology, Australia), and Salil Kanhere (University of New South Wales, Australia)</i>	
Local Model Update for Blockchain Enabled Federated Learning: Approach and Analysis .113.....	
<i>Zhidu Li (Chongqing University of Posts and Telecommunications, China; Advanced Network and Intelligent Interconnection Technology Key Laboratory of Chongqing Education Commission of China; Key Laboratory of Ubiquitous Sensing and Networking in Chongqing, China), Yujie Zhou (Chongqing University of Posts and Telecommunications, China; Advanced Network and Intelligent Interconnection Technology Key Laboratory of Chongqing Education Commission of China; Key Laboratory of Ubiquitous Sensing and Networking in Chongqing, China), Dapeng Wu (Chongqing University of Posts and Telecommunications, China; Advanced Network and Intelligent Interconnection Technology Key Laboratory of Chongqing Education Commission of China; Key Laboratory of Ubiquitous Sensing and Networking in Chongqing, China), and Ruyan Wang (Chongqing University of Posts and Telecommunications, China; Advanced Network and Intelligent Interconnection Technology Key Laboratory of Chongqing Education Commission of China; Key Laboratory of Ubiquitous Sensing and Networking in Chongqing, China)</i>	

Technology Optimizations and Interoperability

PUPoW: A Framework for Designing Blockchains With Practically-Useful-Proof-Of-Work & VanityCoin .122.....	
<i>Yash Chaurasia (International Institute of Information Technology, India), Visvesh Subramanian (International Institute of Information Technology, India), and Sujit Gujar (International Institute of Information Technology, India)</i>	
StreamChain: Building a Low-Latency Permissioned Blockchain For Enterprise Use-Cases .130.....	
<i>Lucas Kuhring (Otto GmbH & Co KG, Germany), Zsolt István (TU Darmstadt, Germany), Alessandro Sorniotti (IBM Research, Switzerland), and Marko Vukolić (Protocol Labs)</i>	
Securely Boosting Chain Growth and Confirmation Speed in PoW Blockchains .140.....	
<i>Ovia Seshadri (Indian Institute of Technology Delhi), Vinay J Ribeiro (Indian Institute of Technology Bombay), and Aditya Kumar (Indian Institute of Technology Delhi)</i>	

InterTrust: Towards an Efficient Blockchain Interoperability Architecture with Trusted Services .150.....	
	<i>Gang Wang (University of Connecticut, USA; University of Connecticut, USA) and Mark Nixon (University of Connecticut, USA)</i>
A Voting-Based Blockchain Interoperability Oracle .160.....	
	<i>Michael Sober (Christian Doppler Laboratory for Blockchain Technologies for the Internet of Things; TU Hamburg, Germany), Giulia Scaffino (Christian Doppler Laboratory for Blockchain Technologies for the Internet of Things; TU Wien, Austria), Christof Spanring (Pantos GmbH, Austria), and Stefan Schulte (Christian Doppler Laboratory for Blockchain Technologies for the Internet of Things; TU Hamburg, Germany)</i>

Insights into Bitcoin and Ethereum

Analysis and Patterns of Unknown Transactions in Bitcoin .170.....	
	<i>Maurantonio Caprolu (Hamad Bin Khalifa University, Qatar), Matteo Pontecorvi (NOKIA Bell Labs, France), Matteo Signorini (NOKIA Bell Labs, France), Carlos Segarra (Imperial College, UK), and Roberto Di Pietro (Hamad Bin Khalifa University, Qatar)</i>
On the Shutdown Price of Blockchain Mining Machines .180.....	
	<i>Shange Fu (Monash University, Australia), Jiangshan Yu (Monash University, Australia), Rafael Dowsley (Monash University, Australia), and Joseph Liu (Monash University, Australia)</i>
Networks of Ethereum Non-Fungible Tokens: A Graph-Based Analysis of the ERC-721 Ecosystem 188	
	<i>S. Casale-Brunet (EPFL SCI STI MM, École Polytechnique Fédérale de Lausanne, Switzerland), P. Ribeca (Biomathematics and Statistics Scotland, United Kingdom), P. Doyle (DegenData.io, PinkSwanTrading Inc., United States of America), and M. Mattavelli (EPFL SCI STI MM, École Polytechnique Fédérale de Lausanne, Switzerland)</i>
Transaction Fees on a Honeymoon: Ethereum's EIP-1559 One Month Later .196.....	
	<i>Daniel Reijsbergen (Singapore University of Technology and Design (SUTD)), Shyam Sridhar (Ethereum Foundation), Barnabé Monnot (Ethereum Foundation), Stefanos Leonardos (Singapore University of Technology and Design (SUTD)), Stratis Skoulakis (Singapore University of Technology and Design (SUTD)), and Georgios Piliouras (Singapore University of Technology and Design (SUTD))</i>
On the Legislation Prospective for Consumer Protection of Stablecoin .205.....	
	<i>Zhiqing Shi (Monash University, Australia), Weiping He (Monash University, Australia), and Joseph Liu (Monash University, Australia)</i>

Short Papers

Identity, Data and Access Control

- Self-Sovereign Identity for the Financial Sector: A Case Study of PayString Service .213.....
*Flaviene Scheidt de Cristo (University of Luxembourg, Luxembourg),
Wazen M. Shbair (University of Luxembourg, Luxembourg), Lucian
Trestioreanu (University of Luxembourg, Luxembourg), Aanchal Malhotra
(Ripple, USA), and Radu State (University of Luxembourg, Luxembourg)*
- DWH-DIM: A Blockchain Based Decentralized Integrity Verification Model for Data Warehouses.221
*Jeroen Bergers (University of Amsterdam, The Netherlands), Zeshun Shi
(University of Amsterdam, The Netherlands), Ken Korsmit (Spatial Eye
B.V., The Netherlands), and Zhiming Zhao (University of Amsterdam, The
Netherlands)*
- Improving Security for Users of Decentralized Exchanges Through Multiparty Computation .229..
Robert Annessi (Nash Exchange) and Ethan Fast (Nash Exchange)
- BlockIoT-RETEL: Blockchain and IoT Based Read-Execute-Transact-Erase-Loop Environment for
Integrating Personal Health Data .237.....
*Manan Shukla (Rensselaer Polytechnic Institute, USA), Jianjing Lin
(Rensselaer Polytechnic Institute, USA), and Oshani Seneviratne
(Rensselaer Polytechnic Institute, USA)*

Smart Contracts and Optimization

- DevOps for Ethereum Blockchain Smart Contracts .244.....
*Maximilian Wöhrer (University of Vienna, Austria) and Uwe Zdun
(University of Vienna, Austria)*
- Evaluating Upgradable Smart Contract .252.....
*Van Cuong Bui (n/a), Sheng Wen (n/a), Jiangshan Yu (n/a), Xin Xia
(n/a), Mohammad Sayad Haghighi (n/a), and Yang Xiang (n/a)*
- A Blockchain-Enabled Anonymous-Yet-Traceable Distributed Key Generation .257.....
*Marina Dehez-Clementi (DISC, ISAE-SUPAERO, France), Jérôme Lacan
(DISC, ISAE-SUPAERO, France), Jean-Christophe Deneuille (ReSCo, ENAC,
France), Hassan Asghar (Macquarie University, Australia), and Dali
Kaafar (Macquarie University, Australia)*
- DAGSec: A Hybrid Distributed Ledger Architecture for the Secure Management of the Internet
of Things .266.....
*Igor D. Alvarenga (Universidade Federal do Rio de Janeiro, Brazil),
Gustavo F. Camilo (Universidade Federal do Rio de Janeiro, Brazil),
Lucas A. C. de Souza (Universidade Federal do Rio de Janeiro, Brazil),
and Otto Carlos M. B. Duarte (Universidade Federal do Rio de Janeiro,
Brazil)*
- Blockchain-Based Scalable Ubiquitous Code Allocation Method Resilient to Congestion .272.....
*Hirotsugu Seike (The University of Tokyo, Japan), Yasukazu Aoki
(Picolab Co., LTD, Japan), and Noboru Koshizuka (The University of
Tokyo, Japan)*

System Applications and Performance

Documenting the Execution of Semantically Modelled Inter-Organisational Workflows on a Distributed Ledger .280.....	
<i>Christoph H.-J. Braun (Institute AIFB, Karlsruhe Institute of Technology (KIT), Germany), Janina Traue (DATEV eG, Germany), Boris Lingl (DATEV eG, Germany), and Tobias Käfer (Institute AIFB, Karlsruhe Institute of Technology (KIT), Germany)</i>	
OpsSC: Decentralized Blockchain Network Operation Workflow for Hyperledger Fabric .287.....	
<i>Tatsuya Sato (Hitachi, Ltd.), Taku Shimosawa (Hitachi, Ltd.), and Yosuke Himura (Hitachi America, Ltd.)</i>	
A Hybrid Blockchain Implementation to Ensure Data Integrity and Interoperability for Public Service Organisations .295.....	
<i>Ali Shahaab (Cardiff Metropolitan University, United Kingdom), Imtiaz Khan (Cardiff Metropolitan University, United Kingdom), Ross Maude (Companies House, United Kingdom), and Chaminda Hewage (Cardiff Metropolitan University, United Kingdom)</i>	
Decentralised Trustworthy Collaborative Intrusion Detection System for IoT .306.....	
<i>Guntur Dharma Putra (UNSW, Sydney; CSCRC, Australia), Volkan Dedeoglu (CSIRO Data61, Australia), Abhinav Pathak (UNSW, Sydney; BITS, Pilani), Salil S. Kanhere (UNSW, Sydney; CSCRC, Australia), and Raja Jurdak (QUT, Brisbane)</i>	
Taming Propagation Delay and Fork Rate in Bitcoin Mining Network .314.....	
<i>Suhan Jiang (Temple University) and Jie Wu (Temple University)</i>	
A Blockchain-Based Approach for Collaborative Formalization of Mathematics and Programs .321	
<i>Jin Xing Lim (Singapore University of Technology and Design (SUTD)), Barnabé Monnot (Ethereum Foundation), Shaowei Lin (Awecom), and Georgios Piliouras (SUTD)</i>	

2021 Workshop on Blockchain Security, Application, and Performance (BSAP-2021)

System Applications (1)

Identifying user Behavior Profiles in Ethereum using Machine Learning Techniques .327.....	
<i>Júlia Almeida Valadares (Federal University of Juiz de Fora, Brazil), Vinicius C. Oliveira (Federal University of Juiz de Fora, Brazil), José Eduardo de Azevedo Sousa (Federal University of Juiz de Fora, Brazil), Heder S. Bernardino (Federal University of Juiz de Fora, Brazil), Saulo Moraes Villela (Federal University of Juiz de Fora, Brazil), Alex Borges Vieira (Federal University of Juiz de Fora, Brazil), and Glauber Gonçalves (Federal University of Piauí, Brazil)</i>	
BATS: A Blockchain-Based Authentication and Trust Management System in Vehicular Networks 333	
<i>Mehrdad Salimitari (Vbit Technologies, Philadelphia), Mohsen Joneidi (University of Central Florida, Orlando), and Yaser P. Fallah (University of Central Florida, Orlando)</i>	

Joint Crypto-Blockchain Scheme for Trust-Enabled CCTV Videos Sharing .341.....	
<i>Mehwish Tahir (Technological University of the Shannon: Midlands Midwest, Ireland), Mamoon Naveed Asghar (Technological University of the Shannon: Midlands Midwest, Ireland), Nadia Kanwal (Lahore College for Women University, Pakistan), Brian Lee (Technological University of the Shannon: Midlands Midwest, Ireland), and Yuansong Qiao (Technological University of the Shannon: Midlands Midwest, Ireland)</i>	
Block-VC: A Blockchain-Based Global Vaccination Certification .347.....	
<i>Alkhansaa A. Abubashim (Temple University, USA; Al-Imam Muhammad Ibn Saud University, Saudi Arabia), Hassan A. Shafei (Temple University, USA; Jazan University, Saudi Arabia), and Chiu C. Tan (Temple University, USA)</i>	

System Applications (2)

SEVA: A Smart Electronic Voting Application using Blockchain Technology .353.....	
<i>Jacob Abegunde (University of Hertfordshire UK), Joseph Spring (University of Hertfordshire UK), and Hannan Xiao (King's College London UK)</i>	
An Examination Protocol for Handling Programmable Answers using a Public Blockchain .361.....	
<i>Shu Takayama (Tokyo Institute of Technology, Japan), Yuto Takei (Tokyo Institute of Technology, Japan), and Kazuyuki Shudo (Tokyo Institute of Technology, Japan)</i>	
CARM: A Blockchain-Based Content Quality Assessment and Rewarding Mechanism .369.....	
<i>Ricardo Giuliano Arcifa (Technological University of the Shannon: Midlands Midwest, Ireland), Enda Fallon (Technological University of the Shannon: Midlands Midwest, Ireland), and Yuansong Qiao (Technological University of the Shannon: Midlands Midwest, Ireland)</i>	
VAULT: A Scalable Blockchain-Based Protocol for Secure Data Access and Collaboration .376.....	
<i>Justin S. Gazsi (Florida International University), Sajia Zafreen (Boise State University), Gaby G. Dagher (Boise State University), and Min Long (Boise State University)</i>	
A Blockchain-Based Contactless Delivery System for Addressing COVID-19 and Other Pandemics 382	
<i>Pratiksha Mittal (Auburn University), Austin Walthall (Auburn University), Pinchen Cui (Auburn University), Anthony Skjellum (University of Tennessee at Chattanooga), and Ujjwal Guin (Auburn University)</i>	

Transactions and DeFi

Diffusion: Analysis of Many-to-Many Transactions in Bitcoin .388.....	
<i>Dylan Eck (Colorado School of Mines), Adam Torek (Boise State University), Steven Cutchin (Boise State University), and Gaby G. Dagher (Boise State University)</i>	
A Peer-To-Peer Ownership-Preserving Data Marketplace .394.....	
<i>Nicolás Serrano (Yachay Tech University, Ecuador) and Fredy Cuenca (Yachay Tech University, Ecuador)</i>	

A Multi-Class Blockchain Services Framework for Communication Networks .401.....	
<i>Gholamreza Ramezan (The University of British Columbia, Canada) and</i>	
<i>Cyril Leung (The University of British Columbia, Canada)</i>	
On Creation of a Stablecoin Based on the Morini's Scheme of Inv&Sav Wallets and Antimoney .409	
<i>Henri T. Heinonen (University of Jyväskylä, Finland)</i>	

Security and Attacks

Chirotonia: A Scalable and Secure e-Voting Framework Based on Blockchains and Linkable Ring Signatures .417.....	
<i>Antonio Russo (IMDEA Networks Institute, Spain), Antonio Fernández</i>	
<i>Anta (IMDEA Networks Institute, Spain), María Isabel González Vasco</i>	
<i>(Universidad Rey Juan Carlos, Spain), and Simon Pietro Romano</i>	
<i>(University of Napoli, Federico II, Italy)</i>	
Towards Malicious Address Identification in Bitcoin .425.....	
<i>Deepesh Chaudhari (CSE, IIT Kanpur, India), Rachit Agarwal (CSE, IIT</i>	
<i>Kanpur, India), and Sandeep Kumar Shukla (CSE, IIT Kanpur, India)</i>	
Knocking on Tangle's Doors: Security Analysis of IOTA Ports .433.....	
<i>Alessandro Brighente (University of Padova, Italy), Mauro Conti</i>	
<i>(University of Padova, Italy), Gulshan Kumar (University of Padova,</i>	
<i>Italy), Reza Ghanbari (Kharazmi University, Iran), and Rahul Saha</i>	
<i>(University of Padova, Italy)</i>	
Analysis of Attack Activities for Honeypots Installation in Ethereum Network .440.....	
<i>Kota Chin (University of Tsukuba, Japan) and Kazumasa Omote</i>	
<i>(University of Tsukuba, Japan, National Institute of Information,</i>	
<i>Japan)</i>	
SECAUCTEE: Securing Auction Smart Contracts using Trusted Execution Environments .448.....	
<i>Harsh Desai (The University of Texas at Dallas) and Murat Kantarcioglu</i>	
<i>(The University of Texas at Dallas)</i>	

Performance Analysis and Improvement

Performance Analysis of a Hyperledger Iroha Blockchain Framework Used in the UK Livestock Industry .456.....	
<i>Konstantinos Ntolkeras (Breedr Ltd, UK), Hossein Sharif (Northumbria</i>	
<i>University, UK), Soroush Dehghan Salmasi (Northumbria University, UK),</i>	
<i>and William Knottenbelt (Imperial College London, UK)</i>	
Improving the Performance of Blockchain Sharding Protocols with Collaborative Transaction Verification .462.....	
<i>Liuyang Ren (University of Waterloo), Paul A. S. Ward (University of</i>	
<i>Waterloo), and Bernard Wong (University of Waterloo)</i>	
Rational Agreement in the Presence of Crash Faults .470.....	
<i>Alejandro Ranchal-Pedrosa (University of Sydney, Australia) and</i>	
<i>Vincent Gramoli (University of Sydney and EPFL, Australia)</i>	

High-Performance Asynchronous Byzantine Fault Tolerance Consensus Protocol .476.....	
<i>Henrik Knudsen (Bouvet, Norway), Jingyue Li (Norwegian University of Science and Technology, Norway), Jakob Svennevik Notland (Norwegian University of Science and Technology, Norway), Peter Halland Haro (Sintef Nord, Norway), and Truls Bakkejord Ræder (Sintef Nord, Norway)</i>	
Containerchain: A Blockchain System Emulator Based on Mininet and Containers .484.....	
<i>Saul Gill (Technological University of the Shannon: Midlands Midwest, Ireland), Brian Lee (Technological University of the Shannon: Midlands Midwest, Ireland), and Yuansong Qiao (Technological University of the Shannon: Midlands Midwest, Ireland)</i>	

Survey and Taxonomy

Blockchain-Based Software Systems: Taxonomy Development .491.....	
<i>Lamia Alashaikh (King Saud University, Saudi Arabia)</i>	
Blockchain Domain-Specific Languages: Survey, Classification, and Comparison .499.....	
<i>Md Tauseef Alam (Indian Institute of Technology Patna, India), Sujit Chowdhury (Indian Institute of Technology Patna, India), Raju Halder (Indian Institute of Technology Patna, India), and Abyayananda Maiti (Indian Institute of Technology Patna, India)</i>	
Blockchain: The Paradox of Consumer Trust in a Trustless System - a Systematic Review .505.....	
<i>Irma Dupuis (The University of Newcastle, Australia), Lisa Toohey (The University of Newcastle, Australia), Sidsel Grimstad (The University of Newcastle, Australia), Berit Follong (The University of Newcastle, Australia), and Tamara Bucher (The University of Newcastle, Australia)</i>	

The International Symposium on Fintech and Blockchain Systems

Block Auction: A General Blockchain Protocol for Privacy-Preserving and Verifiable Periodic Double Auctions .513.....	
<i>Theodoros Constantinides (University of Bristol, UK) and John Cartledge (University of Bristol, UK)</i>	
Effect of Miner Incentive on the Confirmation Time of Bitcoin Transactions .521.....	
<i>Befekadu G. Gebraselase (Norwegian University of Science and Technology, Norway), Bjarne E. Helvik (Norwegian University of Science and Technology, Norway), and Yuming Jiang (Norwegian University of Science and Technology, Norway)</i>	
SAID: ECC-Based Secure Authentication and Incentive Distribution Mechanism for Blockchain-Enabled Data Sharing System .530.....	
<i>Muhammad Rizwan (Southern University of Science and Technology, China), Muhammad Noman Sohail (COMSATS University, Pakistan), Alia Asheralieva (Southern University of Science and Technology, China), Adeel Anjum (COMSATS University, Pakistan), and Pelin Angin (Middle East Technical University, Turkey)</i>	
NFTCert: NFT-Based Certificates With Online Payment Gateway .538.....	
<i>Xiongfei Zhao (University of Macau, Macau) and Yain-Whar Si (University of Macau, Macau)</i>	

The Future of Cryptocurrency Blockchains in the Quantum Era .544	
<i>Sarah Alghamdi (King Fahd University of Petroleum and Minerals, Saudi Arabia) and Sultan Almuhammadi (King Fahd University of Petroleum and Minerals, Saudi Arabia)</i>	
IB2P: An Image-Based Privacy-Preserving Blockchain Model for Financial Services .552	
<i>Zhiyu Xu (Swinburne University of Technology, Australia), Minfeng Qi (Swinburne University of Technology, Australia), Ziyuan Wang (Swinburne University of Technology, Australia), Sheng Wen (Swinburne University of Technology, Australia), Shiping Chen (CSIRO Data61, Australia), and Yang Xiang (Swinburne University of Technology, Australia)</i>	
Decentralized IoT Platform for Flexibility Service Providers in Power Systems .559.....	
<i>Alen Hrga (University of Zagreb, Faculty of Electrical Engineering and Computing, Croatia), Tomislav Capuder (University of Zagreb, Faculty of Electrical Engineering and Computing, Croatia), and Ivana Podnar Žarko (University of Zagreb, Faculty of Electrical Engineering and Computing, Croatia)</i>	
The Blockchain-Based Edge Computing Framework for Privacy-Preserving Federated Learning .566	
<i>Shili Hu (Tongji University, China), Jiangfeng Li (Tongji University, China), Chenxi Zhang (Tongji University, China), Qinpei Zhao (Tongji University, China), and Wei Ye (Suzhou Tongji Blockchain Research Institute, China)</i>	

Blockchain & the Circular Supply Chain

Regulatory Opportunities and Challenges for Blockchain Adoption for Circular Economies .572	
<i>Katrien Steenmans (Coventry University, UK), Phillip Taylor (University of Warwick, UK), and Ine Steenmans (University College London, UK)</i>	
Blockchain's role in Supporting Circular Supply Chains in the Built Environment .578.....	
<i>Qian Li (Cardiff University, United Kingdom) and Yingli Wang (Cardiff University, United Kingdom)</i>	
TRABAC: A Tokenized Role-Attribute Based Access Control using Smart Contract for Supply Chain Applications .584	
<i>Aisyah Ismail (Aglive Lab, Australia), Qian Wu (Aglive Lab, Australia), Mark Toohey (Aglive Lab, Australia), Young Choon Lee (Macquarie University, Australia), Zhongli Dong (The University of Sydney, Australia), and Albert Y. Zomaya (The University of Sydney, Australia)</i>	

IEEE 3rd International Workshop on Advances in Artificial Intelligence for Blockchain (AICChain 2021)

Photrace: A Blockchain-Based Traceability System for Photographs on the Internet .590.....	
<i>Tatsuya Igarashi (Sony Group Corporation, Japan), Takabayashi Kazuhiko (Sony Group Corporation, Japan), Yoshiyuki Kobayashi (Sony Group Corporation, Japan), Hiroshi Kuno (Sony Group Corporation, Japan), and Eric Diehl (Sony Pictures Entertainment Inc., USA)</i>	

Blockchain-Based Mechanism for Robotic Cooperation Through Incentives: Prototype Application in Warehouse Automation .597.....	
<i>Jonathan Grey (DePaul University, USA), Oshani Seneviratne (Rensselaer Polytechnic Institute, USA), and Isuru Godage (DePaul University, USA)</i>	
Author Index 605.	